How one site is ranked: Mysuru, step by step

Every number that takes Mysuru from raw inputs to "#5 of 48 in India, Tier A", with the arithmetic written out. The same formulas run in the atlas page and in the companion Excel workbook (live formulas; change a weight and everything recalculates).

The pipeline in one view

1. **Inputs.** Eight rubric scores, two hazard facts and coordinates for each of 50 candidate cities.
2. **Step A: hard filters.** Drop cities that fail seismic, grid-distance, groundwater or (optionally) cyclone/flood tests. 48 of 50 pass.
3. **Latency.** Distance to four hubs by the haversine formula, converted to milliseconds, averaged.
4. **Step B: normalise.** Rescale every factor to 0–10 using the minimum and maximum among passing cities.
5. **Step C: combine.** Multiply each score by its weight and add (weighted sum), or multiply them as powers (geometric).
6. **Rank and tier.** Sort passing cities; top 30% Tier A, next 40% Tier B, rest Tier C.
7. **Step E: Monte Carlo.** Shake weights and scores by ±20% a thousand times and count how often the site stays near the top.

0 The site and its inputs

Mysuru, Karnataka. District Mysuru, 12.3°N 76.64°E. Chosen because it is a good teaching case: it wins on climate, water, hazards and policy, but is weak on connectivity and demand, so you can see how a strong total is built from uneven parts.

Factor	Raw input	Where it comes from
Power	5	Power: 5 (Report 10 rubric: 50% tariff ₹7.61/kWh, 50% grid and renewables judgement) Report 10 rubric
Connectivity	4	Connectivity: 4 (few carrier-neutral facilities, no cable landing) Report 10 rubric
Latency	11.20 ms	Latency: computed below, not scored by hand computed
Water	7	Water: 7 (Cauvery-fed; safe to semi-critical) Report 10 rubric
Climate	8	Climate: 8 (design 35 °C dry bulb, 24 °C wet bulb, ~2000 free-cooling h/yr) Report 10 rubric
Hazards	9	Hazards: 9 (Zone II) Report 10 rubric
Land	7	Land: 7 Report 10 rubric
Policy	8	Policy: 8 (Karnataka instrument) Report 10 rubric
Demand & talent	3	Demand and talent: 3 (Nascent) Report 10 rubric
Seismic zone	2	IS 1893:2016, zone II reported
220 kV+ substation	15 km	400 kV Mysuru (Kadakola) ~15 km estimate
Groundwater flag	none	Cauvery-fed; safe to semi-critical estimate

The rubric scores are Report 10 judgements on a 0–10 scale (higher is better) and are illustrative, not measurements. Substation names and distances are estimates; no state utility publishes spare capacity.

A Hard filters

A city must pass every test. Failing one removes it before any scoring, so a weak but passing city is never ranked above a city that was excluded. Defaults: zone V excluded, 25 km grid limit, treated water allowed, cyclone/flood filter off.

Test	Mysuru	Limit	Result
Seismic zone below V	2	< 5	**pass**
Nearest 220 kV+ substation	15 km	≤ 25 km	**pass**
Over-exploited groundwater with no treated-water contract	no	—	**pass**
Very-high cyclone or flood exposure (filter off)	no	—	**pass**

Result: **Mysuru stays in.** For contrast, Digha (nearest substation beyond 25 km) and Guwahati (seismic zone V) were removed, so the pool is **48 cities**. That pool size matters in Steps B and the tiers.

L Latency from coordinates

Latency is the one factor computed rather than scored. For each hub, the great-circle distance comes from the haversine formula, then `ms = 0.014 × km` (2 for the round trip, 1.4 for fibre routes that are not straight, ÷ 200 km per millisecond in glass). The model takes the mean over Mumbai, Delhi, Chennai and Bengaluru.

$$a = \sin^2(\Delta\varphi/2) + \cos\varphi_1 \cdot \cos\varphi_2 \cdot \sin^2(\Delta\lambda/2)$$
$$d = 2 \cdot 6371 \cdot \operatorname{asin}(\sqrt{a})$$
$$RTT = 0.014 \times d$$

Hub	Δφ (°)	Δλ (°)	$\sin^2(\Delta\varphi/2)$	$\cos\varphi_1\cos\varphi_2 \sin^2(\Delta\lambda/2)$	a	d (km)	RTT (ms)
Mumbai	6.776	-3.762	0.003492	0.000995	0.004487	854.2	11.96
Delhi	16.314	0.569	0.020132	0.000021	0.020153	1815.0	25.41
Chennai	0.783	3.630	0.000047	0.000955	0.001001	403.3	5.65
Bengaluru	0.672	0.954	0.000034	0.000066	0.000100	127.7	1.79
Mean = (11.96 + 25.41 + 5.65 + 1.79) ÷ 4							**11.20**

Worked line (Bengaluru): Δφ = 0.672°, Δλ = 0.954°. a = 0.000034 + 0.95211 × 0.000069 = 0.000100. d = 12742 × asin(√0.000100) = 127.7 km. RTT = 0.014 × 127.7 = 1.79 ms.

Mysuru's raw latency value is 11.20 ms. The best city in the pool averages 9.73 ms and the worst 20.57 ms (Kolkata).

B Min–max normalisation to a 0–10 scale

Factors come in different units (rubric points, milliseconds), so each is rescaled so the worst passing city scores 0 and the best scores 10:

```
s = 10 × (x − min) ÷ (max − min)          (higher raw is better)
s = 10 × (max − x) ÷ (max − min)          (latency: lower raw is better, so fli
```

Min and max are taken over the 48 cities that passed Step A, so excluding a city can shift everyone's scores.

Factor	x	min	max	Arithmetic	Score s
Power	5	3	9	10 × (5 − 3) ÷ (9 − 3)	**3.33**
Connectivity	4	2	10	10 × (4 − 2) ÷ (10 − 2)	**2.50**
Latency	11.20	9.73	20.57	10 × (20.57 − 11.20) ÷ (20.57 − 9.73)	**8.64**
Water	7	1	7	10 × (7 − 1) ÷ (7 − 1)	**10.00**
Climate	8	1	8	10 × (8 − 1) ÷ (8 − 1)	**10.00**
Hazards	9	3	9	10 × (9 − 3) ÷ (9 − 3)	**10.00**
Land	7	1	10	10 × (7 − 1) ÷ (10 − 1)	**6.67**
Policy	8	3	9	10 × (8 − 3) ÷ (9 − 3)	**8.33**
Demand & talent	3	1	10	10 × (3 − 1) ÷ (10 − 1)	**2.22**

Reading it. Mysuru scores a full 10 on water, climate and hazards because it ties for the best raw score in the pool on each. Its 2.50 on connectivity and 2.22 on demand are low because the pool contains cities (Navi Mumbai scores 10 on both) at the top end. The pool's own range sets the scale, which is why this is a *relative* ranking.

C Weights and the combined score

Weights

The Balanced preset gives raw weights of 25 / 15 / 10 / 15 / 5 / 10 / 5 / 10 / 5 (power, connectivity, latency, water, climate, hazards, land, policy, demand and talent). They sum to 100; each is divided by the sum so the weights add to exactly 1. If you move a slider so they sum to something else, the same division keeps the score on a 0–10 scale.

Method 1: weighted sum (the default)

$$\text{score} = \Sigma\, w_i \times s_i$$

Factor	Score s	Weight w	w × s	Share of total
Power	3.333	0.25	0.8333	13%
Connectivity	2.500	0.15	0.3750	6%
Latency	8.641	0.10	0.8641	14%
Water	10.000	0.15	1.5000	24%
Climate	10.000	0.05	0.5000	8%
Hazards	10.000	0.10	1.0000	16%
Land	6.667	0.05	0.3333	5%
Policy	8.333	0.10	0.8333	13%
Demand & talent	2.222	0.05	0.1111	2%
Total		**1.00**	**6.3502**	

Weighted-sum score for Mysuru = 6.35 out of 10 (Report 10's own published score for it: 6.5).

Power's 0.25 weight times a 3.33 score contributes only 0.83; water and hazards, at a full 10, contribute 1.50 and 1.00. The weak scores cost Mysuru about **3.6** points in total.

Method 2: geometric (a "no weak links" check)

$$\text{score} = \exp\left(\Sigma\, w_i \times \ln(\max(s_i, 0.5))\right) \quad \leftarrow \text{same as } \Pi\, \max(s_i, 0.5)^{w_i}$$

A weighted sum lets one great factor cover a terrible one. The geometric method does not: low scores pull the total down more. The 0.5 floor stops a zero from wiping a city out entirely.

Factor	max(s, 0.5)	ln(...)	w	w × ln
Power	3.333	1.2040	0.25	0.3010
Connectivity	2.500	0.9163	0.15	0.1374
Latency	8.641	2.1565	0.10	0.2157
Water	10.000	2.3026	0.15	0.3454
Climate	10.000	2.3026	0.05	0.1151
Hazards	10.000	2.3026	0.10	0.2303
Land	6.667	1.8971	0.05	0.0949
Policy	8.333	2.1203	0.10	0.2120
Demand & talent	2.222	0.7985	0.05	0.0399
Σ			**1.00**	**1.6917**

Geometric score = exp(1.6917) = 5.43. The gap to 6.35 (0.92 points) is the penalty for Mysuru's uneven profile. Under the geometric method it ranks #7.

R Rank and tier

All 48 passing cities are sorted by score. Tier cut-offs use the pool size:

```
Tier A: ranks 1 to ⌈0.30 × 48⌉ = ⌈14.4⌉ = 15
Tier B: ranks 16 to ⌈0.70 × 48⌉ = ⌈33.6⌉ = 34
Tier C: ranks 35 to 48
```

Rank	City	State	Score
1	Hyderabad periphery	Telangana	7.270
2	Visakhapatnam	Andhra Pradesh	6.643
3	Hyderabad core	Telangana	6.509
4	Jharsuguda	Odisha	6.375
5	**Mysuru**	**Karnataka**	**6.350**
6	Navi Mumbai	Maharashtra	6.343
7	Nava Raipur	Chhattisgarh	6.270
8	Warangal	Telangana	6.227
9	Sri City	Andhra Pradesh	6.191
10	Anantapur	Andhra Pradesh	6.145

Four cities score higher, so Mysuru is #5 of 48, which is inside the top 15: Tier A. Within Karnataka it is #1 of 4: Mysuru 6.35, Hoskote 6.01, Mangaluru 5.69, Bengaluru core 5.52.

The Tier A cut-off sits at a score of 5.99. Mysuru is 0.36 above it, and only 0.007 ahead of #6 Navi Mumbai, so small changes can swap neighbours. That is what Step E tests.

E Monte Carlo: how stable is that rank?

The scores and weights are judgements, so the model asks: "if each were a little different, would Mysuru still do well?" It repeats this 1000 times with random nudges:

1. Draw nine random multipliers from U(0.8, 1.2), one per weight; multiply the raw weights and re-divide by their sum.
2. For each of the 48 cities, draw nine more multipliers, multiply each 0–10 score by its own, and cap at 10.
3. Recompute every city's weighted sum with the nudged inputs and rank them.
4. Record whether Mysuru is in the top 3 of Karnataka and the top 10 of India.

That is 1000 × (9 + 48 × 9) = 441,000 random draws from a fixed-seed generator (seed 20261007, so results repeat). The first three runs, for Mysuru:

Run	Weight multipliers (power … demand)	Mysuru nudged score	India rank	Karnataka rank
1	1.18, 0.97, 0.82, 1.04, 0.90, 0.92, 0.87, 1.12, 0.82	6.03	11	1
2	1.00, 1.11, 0.94, 1.08, 1.16, 1.01, 0.89, 1.02, 0.87	6.20	7	1
3	1.10, 0.84, 0.91, 0.92, 1.16, 0.97, 1.09, 1.10, 1.19	6.08	7	1

Counting over all 1,000 runs:

Karnataka top 3 in 99.8% of runs · India top 10 in 76.1% of runs.

Karnataka has only 4 passing candidates, so "top 3 of the state" is nearly certain (a top-3 place is hard to lose when the state has four cities). The India top-10 share is the more informative figure: Mysuru's rank is solid but not locked, because roughly one run in four pushes it out of the top ten.

What would change the answer?

Same site, same inputs, different weights or method:

Scenario	Combine	Mysuru score	Rank of 48	City ranked #1
Balanced (default)	**Weighted sum**	**6.35**	**5**	**Hyderabad periphery**
Balanced (default)	Geometric	5.43	7	Hyderabad periphery
AI training campus	Weighted sum	6.21	8	Jharsuguda
AI training campus	Geometric	5.41	11	Hyderabad periphery
Interconnection colo	Weighted sum	5.32	23	Navi Mumbai
Interconnection colo	Geometric	4.39	24	Hyderabad periphery
Sustainability-first	**Weighted sum**	**7.58**	**1**	**Mysuru**
Sustainability-first	**Geometric**	**6.70**	**2**	**Hyderabad periphery**
Balanced + report rubric	Weighted sum	6.42	7	Hyderabad periphery

Takeaway. Mysuru's ranking depends on what you value: it is first under Sustainability-first weights and 23rd under Interconnection weights, because its strengths are climate and water and its weakness is connectivity. A site that is rank 5 under one set of weights is not "the fifth best site" in any absolute sense. The ranking tells you which sites are worth a closer look for a given purpose.

Limits to keep in mind

- **Rubric scores are judgements.** Power, connectivity, water, climate, hazards, land, policy and demand are Report 10 ratings, not measured data. Step E partly covers this, but systematic bias (for example, if every Karnataka score were generous) would not show up.
- **Grid headroom is unknown.** Substation distances are estimates; no state utility publishes spare capacity. A load-flow study from the utility is the real test.
- **Relative scale.** Because of min–max, adding or removing a city changes everyone's scores slightly.
- **Latency is a straight-line proxy.** It ignores actual fibre routes and peering; the 1.4 route factor is an assumption.
- **Before shortlisting:** verify block-level groundwater on IN-GRES, check the plot against flood and cyclone maps, and get the real cost of power from the state commission's tariff

order.

Reproduce it

1. Open **dc-site-ranking-workbook.xlsx**, sheet **Walkthrough**. Cell B2 is a drop-down: pick any of the 50 cities and every step above recalculates.
2. On **Settings**, change the weights in row 20 (presets are listed under it), the substation limit, or the filter switches. The **Model** sheet re-ranks all cities.
3. In the atlas, Chapter 10 runs the same engine with sliders and a map. The numbers there match this document: 6.35, rank #5, Tier A.

Prepared from the India Data Centre Atlas ranking model (Report 10, weighted model). Evidence tags: reported = official or published source; estimate = informed estimate; model = computed here.